

Third-Party Political Advertising on Facebook Around the 2022 Australian Federal Election

DATA7201 Data Analytics at Scale — Project Report

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Abstract

Background. The Facebook Ad Library publishes every sponsored political advertisement targeted at Australian users. This report analyses four years of those advertisements (March 2020 to February 2024) to characterise third-party (non-government, non-party) political advertising around the May 2022 Australian federal election.

Methods. Approximately 320 GB of JSON across 4,500 daily snapshots were ingested with PySpark on YARN, deduplicated, and windowed to ± 6 months around election day, reducing the corpus from 40 million to 5.7 million ads. A pandas UDF tagged candidate, party, and government advertising via token-set matching against the Australian Electoral Commission candidate roll. Remaining ads were clustered with Latent Dirichlet Allocation (LDA, $k = 20$) and hand-labelled into thematic categories; commercial clusters were removed, leaving a $\sim 95,000$ -row advocacy corpus. The corpus was analysed in two ways: a deterministic temporal classifier (persistence + centroid) and a hashtag-supervised logistic-regression text classifier.

Findings. Third-party advertising splits into four temporal behaviours — `election_only`, `transitional`, `ongoing`, and `off_campaign`. Election-only advertisers dominate by count and reproduce US PAC-style spending profiles in miniature. Greenpeace Australia dominates the ongoing tier by spend. The content-classifier identifies advertisers whose *messaging* pivoted to electoral themes around polling day even when their *spend* pattern did not, with the Australian Christian Lobby showing the largest content shift.

Conclusions. A combination of temporal and content methodologies is required to cleanly classify third-party political advertising; neither metric alone suffices. The infrastructure choices — PySpark, HDFS, MLlib — were justified by the dataset scale and by the joins required for deduplication and classification.

Word count (excluding abstract, table of contents, tables, figures, captions, references and appendix): *TBC — fill in once body is final.*

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1 Introduction

Winning elections requires reaching voters, and there are many ways to do this. Online advertising is attractive because it allows a message to reach a wide audience with minimal effort. Designing an ad, placing it online, and paying the bill is easier than traditional methods such as door-knocking or physical signage.

Political advertising is not limited to political parties. Third-party groups also participate in election campaigns, and studying digital advertising allows us to examine who these groups are, what they spend, and when they choose to spend it. The Facebook Ad Library API publishes every sponsored political post targeted at Australian users from March 2020 to February 2024, covering three major political events: the COVID-19 pandemic response, the May 2022 federal election, and the October 2023 Voice to Parliament referendum. This report focuses on the 2022 federal election.

At approximately 320 GB of JSON files, the dataset cannot be analysed on a conventional workstation. A single machine cannot hold it in memory, and the joins required for deduplication and classification would be prohibitively slow. Distributed analytics frameworks are the natural fit. PySpark is well suited to this task because it distributes both storage and computation across a cluster, loads data lazily, and its DataFrame API supports efficient analysis across thousands of files [zaharia2016apache, armbrust2015sparksql]. We also leverage Spark MLlib for both the unsupervised topic modelling (LDA) and the supervised text classifier (logistic regression with TF-IDF features) [meng2016mllib].

The investigation was split into three steps: data ingestion and preparation, exploratory analysis, and election-window analysis. These are discussed in the following sections.

2 Dataset Analytics

2.1 Data ingestion and preparation

The data consists of approximately 4,500 JSON files containing around 40 million rows spanning February 2020 to March 2024. Each file is a snapshot of all Australian political advertisements active at that moment, so the same ad appears in multiple files. Each row represents a single ad with around 25 columns covering the ad purchaser, Facebook page, creative text, spend, and impressions. The schema varies over time, with a notable change in June 2022 and another in August 2023; data becomes sparse after August 2023.

A null-fraction analysis was performed to visualise the schema drift. Automatic schema inference collected the superset of all fields across the dataset, and the per-column null fraction by month was plotted as a heatmap (Figure 1). Columns were then modified and merged to produce a consistent schema across the full date range. The numeric fields were stored as nested structs of string-typed lower and upper bounds; these were cast to numeric types and a midpoint was computed for each field, allowing downstream aggregations to work on a single value.

Deduplication was achieved by extracting the snapshot date from each filename and using it to sort snapshots chronologically, grouped by ad ID. A `row_number()` window operation tagged each row with an `ad_seq_no` where 1 indicates the most recent snapshot per ad. Filenames used inconsistent date conventions (some 7-digit, some 8-digit) and required special handling. The cleaned data was windowed to six months either side of the May 2022 federal election and written as Parquet to HDFS, reducing the corpus from approximately 40 million to 5.7 million rows.

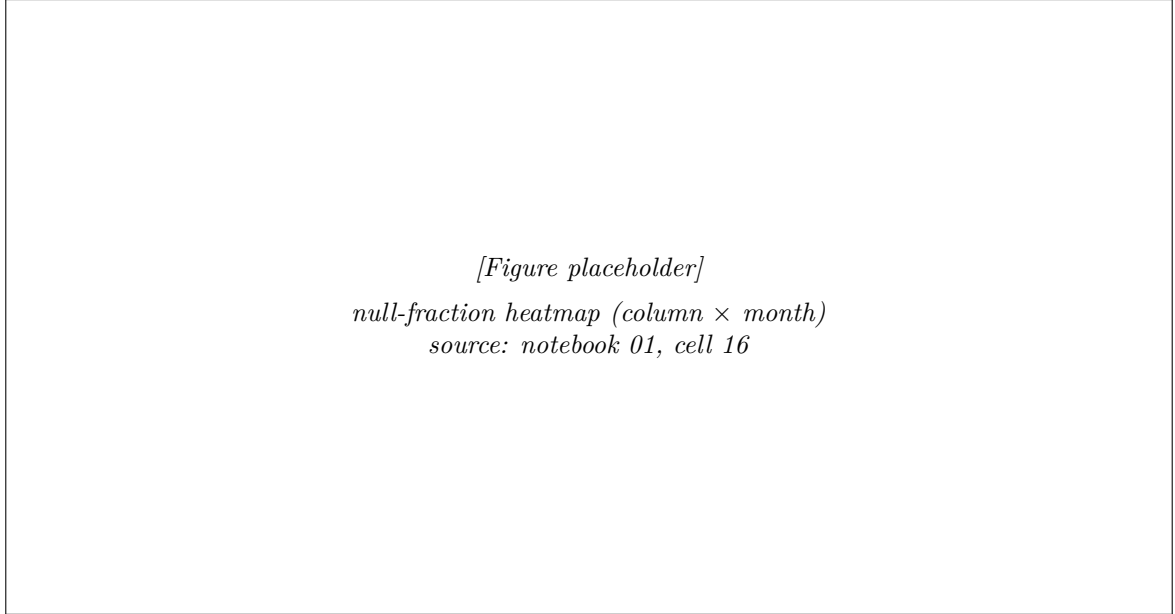


Figure 1: Null-fraction heatmap (column $\times$ month) showing schema drift. The June 2022 transition (singular $\rightarrow$ array fields) and the August 2023 breakage are clearly visible.

2.2 Identifying government advertising

To focus on third-party organisations, official government advertising had to be removed. This included both general political-party advertisements and specific candidate advertisements. The 2022 House of Representatives and Senate candidate rolls were sourced from the Australian Electoral Commission, cleaned, and joined with the main dataset. A straight join was not practical: page names can be misrepresentative, and the byline (the legal “authorised by” field) can be paid by an aide rather than the candidate directly. A pandas UDF was written using Arrow batch transfer between the JVM and Python workers, which gave a 3–10 $\times$ speed-up over a row-by-row Python UDF. The UDF applied three rules in order, first match wins:

1. **Candidate name match.** Each candidate’s name was tokenised to a required token set (surname plus first given name). The UDF flagged a row as a candidate ad if that token set was a subset of either the page-name tokens or the byline tokens. The candidate’s political party was stored in a column.
2. **Party-organisation byline signature.** A lookup table (`aec_parties.csv`) mapped tokenised party names to AEC party codes, e.g. `{liberal, national}` $\rightarrow$ LNP. Signatures were ordered longest-first so `{liberal, national}` matched LNP before LP.
3. **Government byline signature.** A small set of narrow signatures (`{department}`, `{australian, government}`, `{australian, electoral, commission}`) flagged government department advertising. `{commonwealth}` was deliberately excluded due to overlap with the Commonwealth Bank.

Every ad ended up tagged with a `match_type` (`candidate`, `party_org`, `government`, or `null`) and a `political_party` code where applicable (Table 1). Rows with `match_type = null` were the non-government residual required for the third-party analysis.

match_type	total rows	unique ads
null	4,511,742	105,814
party_org	649,857	23,932
candidate	536,232	20,644
government	98,660	1,188

Table 1: Rows and unique ads after government / party / candidate classification.

2.3 Removing commercial noise via LDA

The non-government residual still contained commercial noise. A natural language processing pipeline was developed which used Latent Dirichlet Allocation (LDA) [blei2003lda] to cluster the residual ads into topic groups based on the ad text body. LDA was selected because it is an unsupervised technique — there were no labelled examples to train on — and because it scales naturally to the dataset size when implemented on Spark MLlib. Text was lowercased, regex-tokenised, stop-words removed, and vectorised with a `CountVectorizer` (`minDF=100`, `maxDF=0.3`, `vocabSize=5,000`). After tuning the number of topics on a sample of the data, $k = 20$ gave the cleanest separation. The resulting topics were hand-labelled into a fine-grained `label` (e.g. `anti_woodside_petition`, `wildlife_conservation`) and a coarse `category` (`climate`, `humanitarian_rights`, `political_advocacy`, `cost_of_living`, `noise`). Topics labelled `noise` (commercial, off-topic, non-English) were dropped. The result was a $\sim 95,000$ -row corpus of registered political and political-adjacent advocacy advertising, written partitioned by category for selective downstream reads.

2.4 Temporal analysis: persistence and centroid bands

The goal of the temporal analysis was to characterise how each advertiser’s spending related to the election. Advertisers were ranked by pre-election spend and the top 20 retained for detailed inspection. Spending was aggregated into a weekly time series per advertiser using `date_trunc('week', ad_creation_date)`.

The raw weekly series was bursty. Most advertisers buy media in short pulses, so any summary metric computed on raw weekly spend was heavily leveraged by single-week spikes. A four-week centred rolling mean was applied per advertiser, with a *hard smoothing boundary* placed at election day — the pre- and post-election halves were smoothed independently. This prevented a final pre-election spike from bleeding forward into post-election weeks and softening the cliff edge of advertisers that stopped at the election. The raw series was retained for plotting; only the summary metrics used the smoothed series.

Two metrics were computed per advertiser:

Persistence post-election spend divided by pre-election spend. Captures whether activity continued after the election.

Centroid weighted-mean week from election day. Negative values indicate spending centred before the election; positive values after.

Two metrics were needed because persistence alone produced false positives. A campaign that trickled spend a few weeks past election day, or one that advertised on unrelated topics in the post-election period, both lifted persistence without indicating non-election alignment. Combining persistence with the centroid distinguished these cases.

Advertisers were split into four bands (Table 2):

Band	Rule
election_only	persistence < 0.30 and centroid ≤ 16 weeks
transitional	0.30 ≤ persistence ≤ 0.80
ongoing	persistence > 0.80
off_campaign	persistence < 0.30 and centroid > 16 weeks

Table 2: Four-band classifier applied to top-20 advertisers.

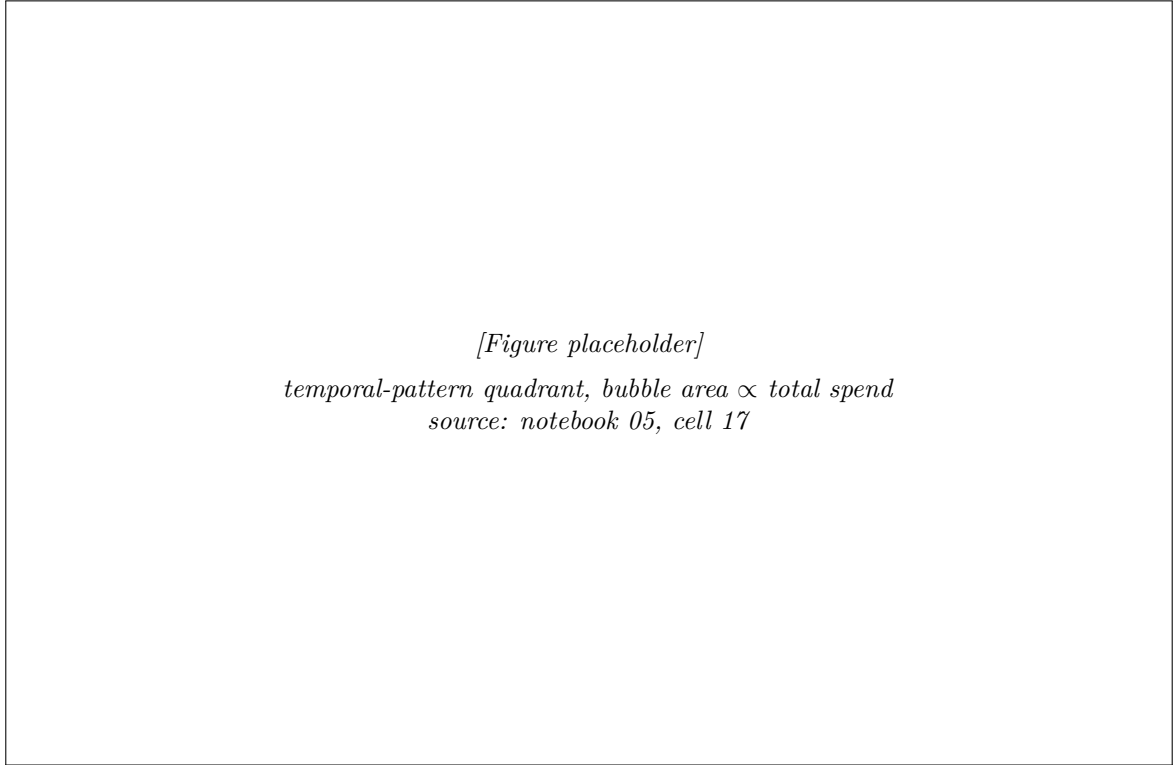


Figure 2: Temporal-pattern quadrant: spend imbalance (x) × centroid timing (y), bubble area ∝ total spend, colour by band. Each quadrant has a clear behavioural interpretation.

2.5 Content analysis: hashtag-supervised classifier

The temporal classifier cannot detect advertisers whose *messaging* pivoted to electoral themes while their *spend* remained steady. To capture this, a content-based classifier was added. We extracted all hashtags from the residual corpus with a regex (`#[A-Za-z0-9_]{2,}`), counted occurrences per band, and hand-curated two reference sets: `hashtags_political.csv` (~210 hashtags such as `#federalelection`, `#morrissonmustgo`, `#bennelong`) and `hashtags_non_political.csv` (~286 hashtags such as `#breakfreefromplastic`, `#architecture`, `#worldrefugeeday`). Each ad’s text was scanned for matches against both sets, yielding two boolean flags. Ads with exactly one flag set were used as training labels:

- Positive: at least one political hashtag and no non-political hashtag
- Negative: at least one non-political hashtag and no political hashtag
- Ambiguous (both flags) or unlabelled (no hashtags): excluded from training, scored at inference

The hashtags themselves were stripped from the text before vectorisation so the model had to learn from the surrounding prose rather than memorising the labels. A logistic-regression classifier was trained over TF-IDF features on the cleaned text. Logistic regression is the standard choice for high-dimensional sparse text classification: tree ensembles underperform on sparse high-dimensional input and yield less interpretable coefficients [joachims1998text, wang2012baselines].

Training set composition: 2,693 positive ads from 273 distinct bylines, 5,262 negative ads from 306 distinct bylines. Class diversity was considerable, which reduced the risk of advertiser-identity leakage. The held-out test AUC was 0.97, accuracy 92%, precision 91%, and recall 87% on the positive class. A leave-one-advertiser-out validation across the top 15 contributors gave an ad-weighted recall of 0.66 — driven down by climate-advocacy bylines (Australian Parents for Climate Action, The Climate Council) whose prose vocabulary overlaps with environmental campaigns in the negative class. Inspection of high-scoring no-hashtag ads showed the model correctly identified political content the upstream filter had missed, including Family First Party, the Preston Market Action Group, and several state-election candidates.

3 Discussion and Conclusions

The non-party, non-government advocacy ecosystem on Facebook around the 2022 election does not behave as a single block. The temporal classifier identifies four behavioural patterns:

The **election_only** band is the largest by count, covering approximately half of the top 20 advertisers. It includes Smart Voting, Solutions for Australia, Advance Australia, the Australian Education Union, GetUp!, It’s Not A Race, Chuffed.org, and Climate 200. These advertisers ramped up before the election and stopped immediately afterwards. Their cumulative spend curves elbow sharply at election day. Keyword analysis confirms this group is dominated by election-specific language — candidate names, party references, specific policies, and direct calls to vote. This is the closest Australian analogue to US PAC-style temporal behaviour, though at smaller scale.

The Australian Cyber Security Centre also appears in this band. It is a government agency and should have been removed by the government filter, but its byline (“Australian Cyber Security Centre”) does not match any of the three government signatures used. This is a known limitation of signature-based government detection.

The **transitional** band contains advertisers that tapered rather than stopping dead — Climate 200, the Australian Conservation Foundation, Thrive By Five, Australia for UNHCR, and the Australian Christian Lobby. The **ongoing** band contains year-round advocates whose post-election rate matches their pre-election rate — Greenpeace Australia, Australian Unions, MSF Australia, and Plan International Australia. Greenpeace Australia dominates this tier by spend (~\$1.5M across the window) and provides the temporal baseline against which everyone else’s election alignment is measured.

The **off_campaign** band catches Shell’s December 2021 spike, which persistence alone would misclassify as election-only. The centroid metric correctly flags it as outside the campaign cycle.

The content classifier identifies a different signal. Among advertisers whose *spend* pattern looked steady or transitional in the temporal analysis, the content classifier reveals that some *messaging* pivoted toward electoral themes around polling day:

- Australian Christian Lobby: content score 0.51 outside the ± 4 -week hot zone, 0.82 inside (shift +0.31)
- Thrive By Five: 0.42 $\rightarrow$ 0.61 (shift +0.19)
- Amnesty International Australia: 0.33 $\rightarrow$ 0.52 (shift +0.19)

Conversely, Greenpeace’s content score *dropped* during the hot zone ($0.15 \rightarrow 0.07$): their pre-election period spending was focused on the Woodside/gas campaign, not on electoral messaging. They maintained their issue focus rather than broadening to election rhetoric. Climate 200 is the inverse case: its content score (0.77) is high year-round, so the hot-zone score (0.81) shows no pivot — they were already maximally political.

The combined methodology is the main contribution of this analysis. Neither metric alone separates the ecosystem cleanly. Temporal banding catches cliff-edge campaigners and off-cycle outliers but cannot detect content pivots in steady spenders. Content classification catches pivots but cannot speak to spend volume or timing concentration. Together, they identify three meaningful patterns:

1. PAC-style election-only campaigners (Smart Voting, Solutions for Australia, Advance Australia) — visible to both methods,
2. year-round advocates that pivot messaging during the campaign (Australian Christian Lobby, Thrive By Five, Amnesty) — visible only to the content method,
3. year-round advocates that stay on issue (Greenpeace, MSF) — visible only to the temporal method.

Lessons learned. Three observations on the infrastructure and methodology choices.

First, the dataset’s scale genuinely justified PySpark. The 40-million-row deduplication via window functions, the LDA fit on the residual corpus, and the pandas UDF for byline matching all leveraged Spark-specific features. None of these would have fit on a workstation in reasonable time.

Second, hand-curation outperformed automation at two key junctures: the LDA topic labels and the hashtag reference sets. Both were short enough (~ 20 topics, ~ 500 hashtags) for human review but provided strong supervision signal for downstream automation.

Third, methodological cross-validation is more informative than tuning a single classifier. The temporal and content classifiers were built from independent signals (spend timing versus ad copy) and their disagreements revealed the most interesting cases.

Main message. The third-party political advertising ecosystem on Facebook around the 2022 Australian federal election is small but structurally rich. It contains both PAC-style election-only operators and year-round advocates, and a meaningful subset of the year-round advocates temporarily broaden their messaging to electoral themes during the campaign window. Detecting that pivot requires content analysis; detecting cliff-edge spending requires temporal analysis. Combining the two gives a complete picture.

A Code

The full analysis pipeline is implemented in six Jupyter notebooks at https://github.com/<USER>/FB_API (replace with the actual URL). Each notebook is self-contained and writes its output as Parquet on HDFS or as CSV in the repository’s `data/` directory.

01_data_loading.ipynb Raw JSON ingestion, schema flattening, deduplication, window filtering, write of v1 Parquet.

02_preprocessing.ipynb Candidate, party, and government byline matching via pandas UDF; write of v2 Parquet.

03_topics.ipynb LDA topic modelling, k sweep, topic inspection, output of `topic_terms.csv` for hand-labelling.

04_topic_join.ipynb Join of hand-labelled topics onto v2, filter to political-adjacent residual, write of v3 Parquet.

04b_hashtag_analysis.ipynb Regex extraction of hashtags across v3, count by band, export of data/hashtags.csv.

05_election_spenders.ipynb Top-20 ranking, weekly aggregation, election-boundary smoothing, persistence + centroid computation, four-band classification, export of band CSVs and visualisations.

06_election_content_classifier.ipynb Hashtag-supervised logistic-regression text classifier; FP / FN / LOO validation; per-advertiser and per-week scoring; pivot-detection table.

Selected code excerpts follow.

Election-boundary smoothing (nb 05)

```
# Hard smoothing boundary at election day --- pre and post halves
# smoothed independently so cliff-edged advertisers do not bleed forward.
weekly['is_post'] = weekly['week'] >= ELECTION_DATE
weekly['spend_smooth'] = (weekly
    .groupby(['bylines', 'is_post'])['spend']
    .transform(lambda s: s.rolling(window=4, center=True, min_periods=1).mean())
    )
```

Hashtag-supervised labelling (nb 06)

```
pol_re = r'#(' + '|'.join(re.escape(t) for t in political_hashtags) + r')\b'
non_re = r'#(' + '|'.join(re.escape(t) for t in non_political_hashtags) + r')\b'

v3 = v3.withColumn('has_political_hashtag', col('raw_text').rlike(pol_re))
v3 = v3.withColumn('has_non_political_hashtag', col('raw_text').rlike(non_re))

# Strip hashtags before vectorising --- model must learn from surrounding prose.
v3 = v3.withColumn('text', regexp_replace(col('raw_text'), r'#\w+', ' '))

labelled = v3.withColumn('label',
    when( col('has_political_hashtag') & ~col('has_non_political_hashtag'), 1.0)
    .when(~col('has_political_hashtag') & col('has_non_political_hashtag'),
        0.0)
    .otherwise(None))
```